



8873/01

17 September 2018

1 hour

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

Write in soft pencil.
Do not use staples, paper clips, glue or correction fluid.
Write your name, PDG and NRIC / FIN number on the Answer Sheet in the spaces provided.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the Multiple Choice Answer Sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

Multiple Choice Answer Sheet

Write your name, PDG and NRIC / FIN number, **including** the reference letter.

Shade the NRIC / FIN number.

Exam Title: JC2 PRELIM

Exam Details: H1 Chemistry / Paper 1

Date: 17/09/2018

This document consists of **11** printed pages.

- 1 An unknown gaseous hydrocarbon C_xH_y of volume V is mixed with oxygen and then completely combusted.

Which volume of oxygen reacts with the hydrocarbon?

- A $(\frac{x}{2} + \frac{y}{4})V$ B xV C $(x + \frac{y}{2})V$ D $(x + \frac{y}{4})V$

- 2 A sample of silicon contains three naturally occurring isotopes, ^{28}Si , ^{29}Si and ^{30}Si .

The sample is made up of 92.23% ^{28}Si and the relative atomic mass of silicon in this sample is 28.10.

What is the percentage of the isotope ^{29}Si in the sample?

- A 2.23% B 3.89% C 5.54% D 7.77%

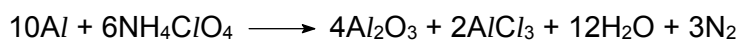
- 3 *Use of the Data Booklet is relevant to this question.*

How many hydrogen atoms are present in 15 g of ethanoic acid?

[L = Avogadro constant]

- A $\frac{1}{L}$ B $\frac{L}{4}$ C L D $60L$

- 4 The space shuttle's upward thrust, on lift off, came from the reaction between aluminium and ammonium perchlorate.



Which statement about the overall reaction is correct?

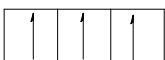
- 1 Aluminium is oxidised.
- 2 Chlorine is reduced.
- 3 Hydrogen is oxidised.
- 4 Nitrogen is reduced.
- 5 Oxygen is oxidised.

- A 1 and 2 only
 B 2 and 5 only
 C 1, 2 and 3 only
 D 3, 4 and 5 only

5 Which of the following statements about isotopes is correct?

- A They have the same number of protons but different number of electrons.
- B They have the same number of protons but different number of neutrons.
- C They have the same number of neutrons but different number of electrons.
- D They have the same number of neutrons but different number of protons.

6 Which diagram represents the arrangement of electrons in the 3rd and 4th quantum shell of a Ca³⁺ ion?

- A**
- | | | |
|---|---|---|
|  |  |  |
| 3s | 3p | 4s |
- B**
- | | | |
|---|---|---|
|  |  |  |
| 3s | 3p | 4s |
- C**
- | | | |
|---|---|---|
|  |  |  |
| 3s | 3p | 4s |
- D**
- | | | |
|---|---|---|
|  |  |  |
| 3s | 3p | 4s |

7 Use of the Data Booklet is relevant to this question.

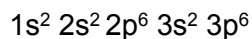
The successive ionisation energies, in kJ mol⁻¹, of an element **D** are given below.

900 1730 2570 3830 4860 6950 8210 12600

What is **D**?

- A** $_{52}\text{Te}$ **B** $_{53}\text{I}$ **C** $_{84}\text{Po}$ **D** $_{85}\text{At}$

8 Argon has the following electronic configuration:



Which of the following ions does **not** have the same electronic configuration as argon?

- A** P^{3-} **B** Cl^- **C** Ca^+ **D** Ti^{4+}

9 Which of the following molecules has the greatest polarity?

- A** BF_3 **B** NCl_3 **C** SiCl_4 **D** SF_6

10 As a fixed mass of ice melts at 0 °C, which of the following statements are true?

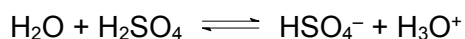
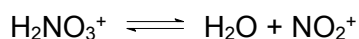
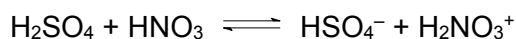
- 1 The weak covalent bonds between H₂O molecules are broken.
 - 2 There is a contraction in volume.
 - 3 The temperature of the ice–water mixture increases.
 - 4 The average number of hydrogen bonds per H₂O molecule remains unchanged.
- A** 1 and 2 only
B 3 and 4 only
C 2 and 3 only
D 2 and 4 only

11 In which pair of species are the values of the bond angles the most different?

- A** PF₃ and NH₃
B NH₄⁺ and SO₄²⁻
C NH₂⁻ and H₂S
D CH₃⁻ and CH₃⁺

12 The Brønsted–Lowry theory describes acid and base character.

When concentrated sulfuric acid and concentrated nitric acid are mixed, the following reactions occur.



Which species are bases in these reactions?

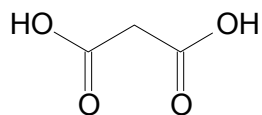
- 1 HSO₄⁻
 - 2 HNO₃
 - 3 NO₂⁺
- A** 1 only
B 2 only
C 3 only
D 1 and 2 only

- 13 A solution was made by mixing equal volumes of two acids of pH 1 and pH 3.

What is the pH of the resulting solution?

- A 1.0 B 1.3 C 2.0 D 2.5

- 14 Malonic acid is a weak dibasic acid and has the structure shown below.

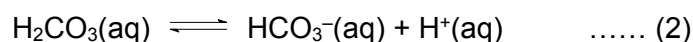
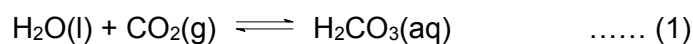


The concentration of a solution of malonic acid was determined by titrating it against a standard solution of aqueous sodium hydroxide in a burette. The pH of the first and second equivalence points are found to be 4.5 and 9.1 respectively.

Which of the following is a suitable indicator for part of this titration?

	<i>indicator</i>	<i>pH range</i>
A	thymol blue	1.2 to 2.8
B	methyl orange	3.2 to 4.4
C	bromocresol green	3.8 to 5.5
D	phenol red	6.4 to 8.0

- 15 The main buffering agent to maintain the blood pH at 7.4 is the $\text{H}_2\text{CO}_3/\text{HCO}_3^-$ system. The following two equations show the equilibria of $\text{H}_2\text{CO}_3/\text{HCO}_3^-$ system in the body.



During strenuous exercise, the blood pH may fall below 7.4.

Which of the following actions will help the body increase the blood pH?

- 1 Breathing into a paper bag.
- 2 Removing H^+ ions from the blood.
- 3 Secreting HCO_3^- ions into the blood.

- A 1 and 2 only B 2 and 3 only C 1 and 3 only D 1, 2 and 3

16 Which series is correctly arranged in order of increasing values?

- A** atomic radius of P, S, Cl
- B** ionic radius of Na^+ , Mg^{2+} , Al^{3+}
- C** melting point of Al, Si, P
- D** second ionisation energy of Si, P, S

17 An element X in Period 3 is a semiconductor and has a chloride which reacts with water to form an acidic solution.

An element Y has an atomic number one less than element X.

What is a property of the oxide of element Y?

- A** It is a gas at room temperature.
- B** It is amphoteric.
- C** It is covalent.
- D** Its formula is YO_2 .

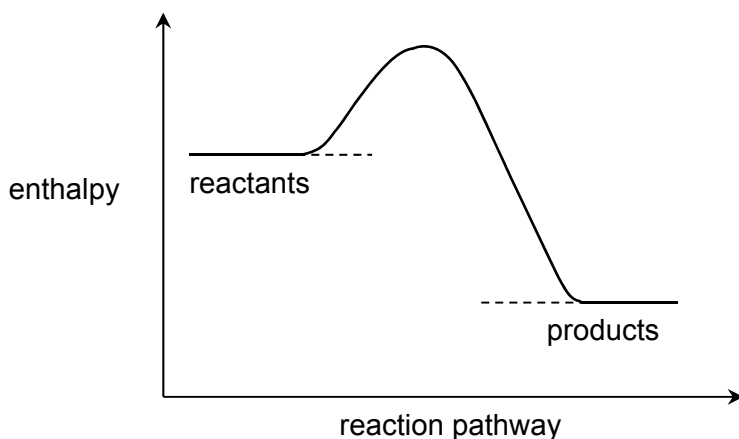
18 The information relates to element Z.

- Z is in Period 3 of the Periodic Table.
- Z has a lower electrical conductivity than Mg.
- A Z atom has a half-filled subshell in its ground state.
- Z forms an acidic oxide on exposure to air.

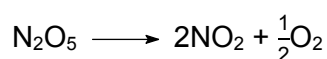
What is Z?

- | | | | |
|-------------|-------------|------------|-------------|
| A Na | B Si | C P | D Cl |
|-------------|-------------|------------|-------------|

- 19 Which enthalpy change could **never** be correctly represented by the following enthalpy diagram?



- A standard enthalpy change of combustion
 B standard enthalpy change of formation
 C standard enthalpy change of neutralisation
 D standard enthalpy change of vapourisation
- 20 Which statement best defines the term bond energy for a diatomic gaseous molecule PQ?
- A energy released when one mole of PQ is formed from its gaseous ions
 B energy released when one molecule of PQ is formed from its gaseous atoms
 C energy required when one molecule of PQ is broken into its gaseous atoms
 D energy required when one mole of PQ is broken into its gaseous atoms
- 21 N_2O_5 decomposes according to the equation.

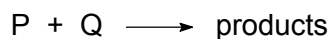


The decomposition is first order and the half-life of N_2O_5 is 150 s.

When a sample containing 0.10 mol of N_2O_5 was placed in a 1 dm³ vessel, what is the concentration of NO_2 after 5 minutes?

- A 0.150 mol dm⁻³
 B 0.100 mol dm⁻³
 C 0.050 mol dm⁻³
 D 0.025 mol dm⁻³

- 22 The following reaction is first order with respect to P and zero order with respect to Q.

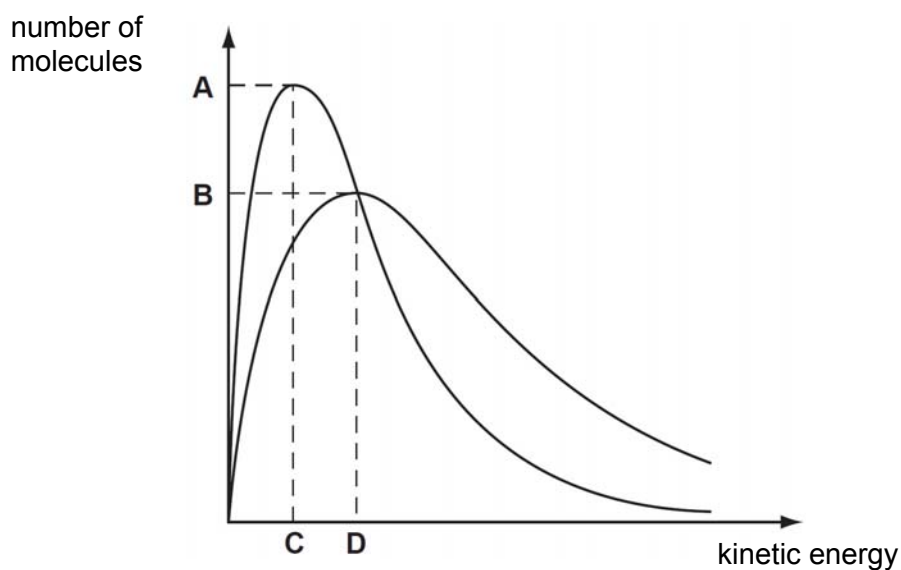


If the rate constant doubles for each 10 °C rise in temperature, which of the following sets of conditions will give the greatest rate of reaction?

	[P] / mol dm ⁻³	[Q] / mol dm ⁻³	T / °C
A	0.1	0.2	40
B	0.1	0.3	30
C	0.2	0.2	30
D	0.3	0.1	20

- 23 The diagram shows the Maxwell–Boltzmann energy distribution curves for molecules of a sample of a gas at two different temperatures.

Which letter on the axes represents the proportion of molecules with the most probable energy at the lower temperature?

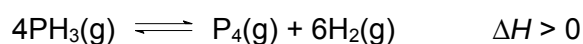


- 24 Metals such as rhodium (Rh), palladium (Pd) and platinum (Pt) are surface catalysts used in the catalytic converter to convert unburnt hydrocarbon, CO and oxides of nitrogen to less harmful gases such as CO₂, H₂O and N₂ to reduce air pollution.

Which of the following statements are correct?

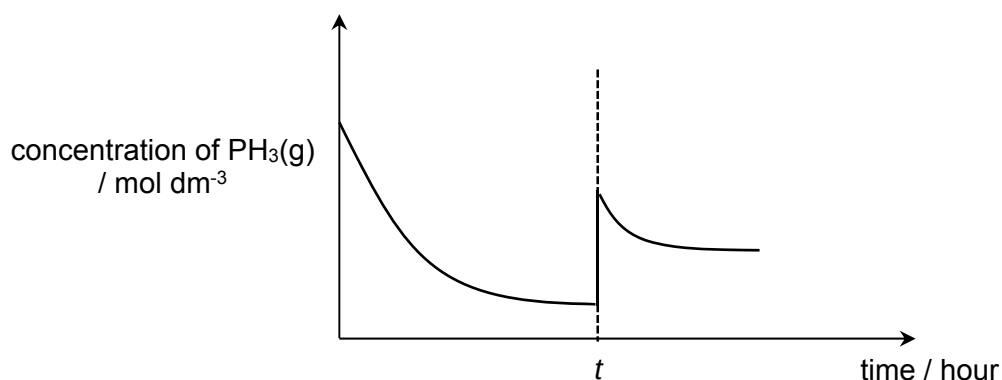
- 1 The metals increase the rate of reaction by providing an alternative pathway of lower activation energy and make the reaction more exothermic.
 - 2 The metals act as heterogeneous catalysts.
 - 3 The metals in the catalytic converter will be depleted with time and needs to be replaced.
 - 4 Rh, Pd and Pt are inert metals with high melting points.
- A 1 and 3 only
 B 2 and 4 only
 C 2 only
 D 3 only

- 25 Phosphine, PH₃, decomposes to give phosphorus and hydrogen gas.



The graph below shows the change in concentration of PH₃ over time until the reaction mixture reaches equilibrium at a constant temperature of 400 K.

Which of the following are **not** possible changes made at t hour?



- 1 Addition of PH₃ to the reaction mixture
 - 2 Reduction in the volume of the vessel
 - 3 Removal of P₄ from the reaction mixture
 - 4 Addition of a catalyst
- A 1 and 2 only B 3 and 4 only C 2 and 4 only D 2, 3 and 4 only

26 Which of the following statements about graphene is true?

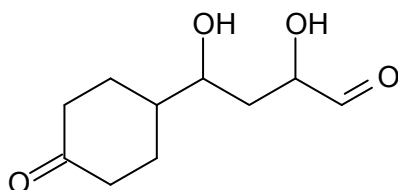
- A Each carbon atoms form σ bonds with 6 neighbouring carbon atoms.
- B It is an extremely thin 3-dimensional layer of carbon atoms.
- C It is a good thermal conductor.
- D Weak instantaneous dipole-induced dipole forces of attraction exist between the layers of graphene.

27 "Clearfilm" is manufactured from a polymer made by copolymerizing $\text{CH}_2=\text{CHCl}$ with $\text{CH}_2=\text{CCl}_2$ in a regular "head to tail" linkage where CH_2 is taken as the "head" of the monomer.

Which of the following could represent part of the polymer chain in "clearfilm"?

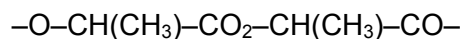
- A $-\text{CHCl}-\text{CH}_2-\text{CCl}_2-$
- B $-\text{CCl}_2-\text{CCl}_2-\text{CH}_2-\text{CHCl}-$
- C $-\text{CH}_2-\text{CHCl}-\text{CCl}_2-$
- D $-\text{CH}_2-\text{CCl}_2-\text{CHCl}-\text{CH}_2-$

28 Which of the following statements are true about the following compound?



- 1 Its molecular formula is $\text{C}_{10}\text{H}_{14}\text{O}_4$.
 - 2 It reacts with sodium hydrogen carbonate to give carbon dioxide gas.
 - 3 It reacts with hot acidified potassium dichromate to form a product with 5 O atoms.
 - 4 It reacts with 2 mol of sodium metal to form 1 mol of hydrogen gas.
- A 1 and 2 only
 - B 3 and 4 only
 - C 2 and 4 only
 - D 1 and 3 only

29 Part of the structure of a polyester is shown below:

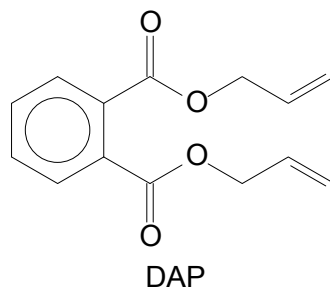


Which of the following statements are true?

- 1 It is a thermoplastic.
- 2 The polymer chains are held together by hydrogen bonds.
- 3 The C–O–C angle in the polymer is 120°.
- 4 It undergoes acid hydrolysis to give $\text{HOCH}(\text{CH}_3)\text{CO}_2\text{H}$.

- A** 1 and 2 only
B 1 and 4 only
C 2 and 3 only
D 3 and 4 only

30 Diallyl phthalate (DAP) has a structure shown below. As a monomer, it is used as a cross-linking agent in unsaturated polyesters. When it undergoes polymerisation to form poly(diallyl phthalate), one of the two $>\text{C}=\text{C}<$ is first used to join up the DAP monomers into a pre-polymer. During the molding process, the second $>\text{C}=\text{C}<$ polymerises leading to a highly cross-linked thermoset polymer.



Its polymer is used in the production of thermosetting molding powders.

Which of the following statements is **not** true?

- A** Cross-linkages with unsaturated polyesters are formed by the unsaturated $>\text{C}=\text{C}<$ bonds of DAP monomers.
- B** Poly(diallyl phthalate) is soft and flexible.
- C** Poly(diallyl phthalate) is an electrical insulator.
- D** Poly(diallyl phthalate) is resistant to high temperature.

<i>Question Number</i>	<i>Key</i>	<i>Question Number</i>	<i>Key</i>
1	D	16	D
2	C	17	B
3	C	18	C
4	A	19	D
5	B	20	D
6	D	21	A
7	D	22	A
8	C	23	A
9	B	24	B
10	D	25	D
11	D	26	C
12	D	27	A
13	B	28	B
14	C	29	B
15	B	30	B

A	5
B	9
C	6
D	10